

Spatial and temporal patterns of the macroinvertebrate community in Tonga Lake (Algeria) in relation to water quality

Padrões espaciais e temporais da comunidade de macroinvertebrados no lago Tonga (Argélia) em relação à qualidade da água

Patrones espaciales y temporales de la comunidad de macroinvertebrados del lago Tonga (Argelia) en relación con la calidad del agua

DOI: 10.34188/bjaerv7n4-079

Submetido: 01/08/2024

Aprovado: 30/09/2024

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ABSTRACT

The current study aimed to investigate the macroinvertebrate community and its relationship with environmental parameters in a protected lake, Tonga Lake (Northeast Algeria). Macroinvertebrate sampling was carried out at three sites at Tonga Lake over one year (March 2017 to February 2018). This lake located in the El Kala National Park, has been a Ramsar classified area since 1983. A total of 42,849 individuals of macroinvertebrates were collected. The thirty species (30) identified belong to 20 families: Culicidae; Corexidae; Microvelinae; Belostomatidae; Naucoridae; Nepidae; Notonectidae; Pleidae; Eylidae; Pionidae; Arrenuridae; Hydrachnidae; Daphnidae; Cyclopidae; Cyprididae; Baetidae; Leptophlebiidae; Libellulidae; Aeshnidae and Lestidae. The highest diversity was recorded during the summer. Seven physicochemical parameters were measured at each site: pH, dissolved oxygen, temperature, nitrite, nitrate, ammonium, and orthophosphate. The analysis verified by the Kruskal Wallis test between the means of inter-site and inter-season comparisons shows that the change in physicochemical parameters was only significant for the temperature between seasons. The rest of the water parameters measured revealed non-significant variations. Overall, the kinetic abundance of the different families and the distribution of species change according to the season and the site.

Keywords: biodiversity, aquatic insects, inventory, freshwater, Algeria.

RESUMO

O presente estudo teve como objetivo investigar a comunidade de macroinvertebrados e a sua relação com os parâmetros ambientais num lago protegido, o Lago Tonga (Nordeste da Argélia). A amostragem de macroinvertebrados foi realizada em três locais no lago Tonga durante um ano (março de 2017 a fevereiro de 2018). Este lago, situado no Parque Nacional de El Kala, é uma zona classificada como Ramsar desde 1983. Um total de 42.849 indivíduos de macroinvertebrados foram recolhidos. As trinta espécies (30) identificadas pertencem a 20 famílias: Culicidae; Corexidae; Microveliinae; Belostomatidae; Naucoridae; Nepidae; Notonectidae; Pleidae; Eylaidae; Pionidae; Arrenuridae; Hydrachnidae; Daphnidae; Cyclopidae; Cyprididae; Baetidae; Leptophlebiidae; Libellulidae; Aeshnidae e Lestidae. A maior diversidade foi registada durante o verão. Foram medidos sete parâmetros físico-químicos em cada local: pH, oxigénio dissolvido, temperatura, nitrito, nitrato, amónio e ortofosfato. A análise verificada pelo teste de Kruskal Wallis entre as médias das comparações entre locais e entre estações mostra que a variação dos parâmetros físico-químicos só foi significativa para a temperatura entre estações. Os restantes parâmetros da água medidos revelaram variações não significativas. Globalmente, a abundância cinética das diferentes famílias e a distribuição das espécies variam consoante a estação e o local.

Palavras-chave: biodiversidade, insectos aquáticos, inventário, água doce, Argélia.

RESUMEN

El presente estudio tuvo como objetivo investigar la comunidad de macroinvertebrados y su relación con los parámetros ambientales en un lago protegido, el lago Tonga (noreste de Argelia). El muestreo de macroinvertebrados se llevó a cabo en tres sitios del lago Tonga durante un año (de marzo de 2017 a febrero de 2018). Este lago situado en el Parque Nacional de El Kala, es una zona clasificada Ramsar desde 1983. Se recogieron un total de 42.849 individuos de macroinvertebrados. Las treinta especies (30) identificadas pertenecen a 20 familias: Culicidae; Corexidae; Microveliinae; Belostomatidae; Naucoridae; Nepidae; Notonectidae; Pleidae; Eylaidae; Pionidae; Arrenuridae; Hydrachnidae; Daphnidae; Cyclopidae; Cyprididae; Baetidae; Leptophlebiidae; Libellulidae; Aeshnidae y Lestidae. La mayor diversidad se registró durante el verano. En cada sitio se midieron siete parámetros fisicoquímicos: pH, oxígeno disuelto, temperatura, nitrito, nitrato, amonio y ortofosfato. El análisis verificado mediante la prueba de Kruskal Wallis entre las medias de las comparaciones entre sitios y entre estaciones muestra que el cambio en los parámetros fisicoquímicos sólo fue significativo para la temperatura entre estaciones. El resto de los parámetros del agua medidos revelaron variaciones no significativas. En general, la abundancia cinética de las distintas familias y la distribución de las especies cambian según la estación y el lugar.

Palabras clave: biodiversidad, insectos acuáticos, inventario, agua dulce, Argelia.

1 INTRODUCTION

The Mediterranean basin has been described as one of the richest and most complex regions in terms of geology, biology, and culture (Blonde et al., 2010; Plieninger et al., 2023). Its high biological diversity and degree of endemism make it one of the world's 34 'hot spots' (Myers et al., 2000). Algeria currently has 2 375 wetlands, including 50 Ramsar sites of international importance (DGF, 2017). In the north-eastern part of Algeria, one of the country's most heavily irrigated areas lies the El Kala wetland complex, one of the most important in North Africa and the Mediterranean basin due to its position on migration routes and its high biodiversity (Amara Korba et al., 2016; Djamaï et al., 2019), characterized by the presence of marine and lacustrine ecosystems, including Lake Tonga, the site of our study. The integral reserve of Lake Tonga was designated as a Ramsar site in 1983 and is located in the UNESCO biosphere reserve (Mahmoudi, Bendali-Saoudi, and

Soltani, 2023). It is a seasonal freshwater lake, linked to the Mediterranean and provides extensive habitat for aquatic plants and nesting sites for rare and threatened waterbirds (Toubal et al., 2014).

Continental aquatic ecosystems cover 0.8% of the planet's total surface area and are home to at least 6% of the species described to date. However, aquatic biodiversity is declining worldwide. The composition of a site's biotic community reflects all the factors that influence the environment, including macroaquatic invertebrates. The latter is poorly known or ignored, yet they constitute an essential group, particularly in the transformation of organic matter, fish and bird food, and their potential interest as a diagnostic element in the pollution of aquatic sites (Khedimallah and Tadjine, 2016). Despite their many piscicultural and medical benefits, the insects of African aquatic environments are unfortunately poorly known; in particular (Sanogo, Kabre, and Cecchi, 2014; Mahmoud and Riad, 2020).

The use of any biological community in water conservation and/or monitoring systems requires the characterization of its diversity and structure (Allan and Johnson, 1997). The identification of species most often requires an understanding of the diversity of arthropods as a whole and reasoning at the community level. This means carrying out inventories on a more or less large scale. Wetlands contribute significantly to regional freshwater biodiversity and provide important ecosystem services (Biggs, Fumetti, and Kelly-Quinn, 2017; Bouchaala et al., 2019; Fartas et al., 2024). They are also frequently impacted by various anthropogenic activities notably following the growing use of fertilizers (Halaimia, Tine, Tine-Djebbar, and Soltani, 2021), and pesticides (Hamida et al., 2021). The human pressure was reported to alter the diversity and composition of the macroinvertebrate community (Stamenković et al., 2022). Moreover, increasing evidence demonstrated that climate change has affected wetlands via particularly a reduction in precipitation patterns. Despite their importance, wetlands are still studied in Algeria. In this context, recent studies were conducted in Northeast Algeria to perform inventories of macroinvertebrates in various aquatic ecosystems like Burgas Lake (Benmalelk, Bendali-Saoudi, and Soltani, 2018), Seybouse River (Baaloudj et al., 2020), Réghaïa Lake (Djitli et al., 2021), in an archaeological site Roman Ruins of Hippone (Bendjedid, Bendali-Saoudi, and Soltani, 2023), at Bouhamdane River (Rouibi et al., 2021), Fetzara Lake (Boulares et al., 2023), Echatt Lake (Houmani, Bendali-Saoudi, and Soltani, 2023), at Hauts Plateaux (Chaib et al., 2023).

The present study aims to remedy this shortcoming by providing an initial database on the macrofauna of aquatic invertebrates in Tonga Lake. Its objectives are 1/ to assess a qualitative and quantitative inventory of macrofauna collected from three study sites over one year and 2/ to identify the biotic and abiotic parameters structuring these populations in this particular aquatic ecosystem. The attempt data contribute to our understanding of the responses of freshwater macroinvertebrates

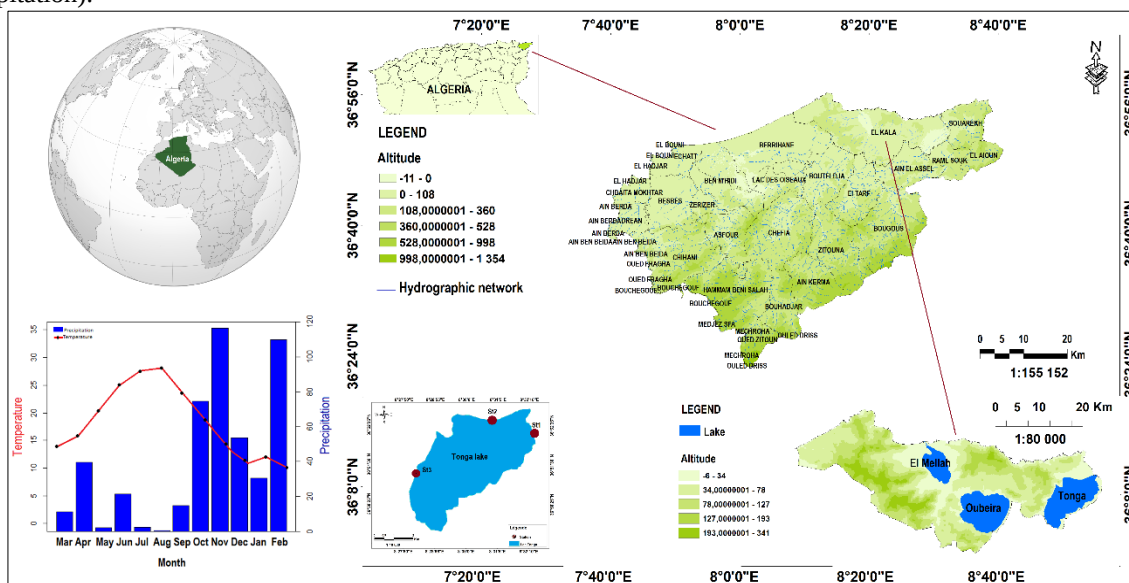
from a particular aquatic ecosystem to multiple anthropogenic pressures.

2 MATERIALS AND METHODS

2.1 STUDY SITE

Lake Tonga is located at 36°49' N-8°32' E in the extreme north-east of the El Kala National Park (state of El Tarf) and Algeria (Figure 1). It lies to the east of the town of El Kala, 70 km east of the town of Annaba (Loucif et al., 2020). To the north, Tonga is bordered by coastal dunes separating it from the sea. It is a palustrine freshwater lake in communication with the Mediterranean Sea via an artificial canal, the Messida Canal (Djamai et al., 2019). Its average altitude is 2.20 meters and its average depth varies from 1 to 1.2 m (Ouarti et al., 2023). This study involved sampling at 3 sites, located on the periphery of the lake. The sites were selected based on of water accessibility criteria, their remoteness, and their geographical position (Mahmoudi et al., 2023). They are as follows: (S1: 36° 87' N- 8° 53' E), (S2: 36° 88' N- 8° 52' E) and (S3: 36° 86' N 8°47' E) (Figure 1).

Figure. 1: Geographical location of the Tonga lake and sampling sites and meteorological curves (Temperature and Precipitation).



Source: Prepared by the authors (2024).

2.2 SAMPLING AND IDENTIFICATION OF INVERTEBRATE FAUNA

The monthly samples were taken over 12 months (March 2017 - February 2018) at three selected sites. The sampling was carried out using a 0.5 mm mesh vacuum net at a depth of 30 to 50 cm, using the Dipping method. The invertebrate macrofauna collected was placed in containers filled with water, labeled and immediately transported to the laboratory and processed within max 5 h from collection time. The specimens captured were first sorted into morpho-species and preserved in 70% (Mahmoudi et al., 2023). In the simplest case, the individuals in the population

can be counted. However, given the high density and abundance of individuals, it will be difficult to count the population accurately. To do this, we estimated the sampled population of Cladocerans and Copepods. After homogenizing 1000 ml (volume p) of water from the bed, 20 ml (volume a) was taken three times. However, the individuals were counted according to the formula of Dreux (1963):

- N) The estimated population, of unknown size;
- n) The mean value of the individuals sampled from the 3 replicates a;
- p) Volume comprising the population to be estimated.

$$N = \frac{np}{a}$$

The systematic determination was carried out using appropriate dichotomous keys, based on a series of specific morphological characters and structures. Taxa are identified using a binocular magnifying glass (ZEISS) and a photonic microscope (Leica DM500). The identification of specimens is carried out on a slide, in a drop of glycerine after rinsing with distilled water. Generic and specific identification of specimens was made for Culicidae using the software developed by Schaffner et al, (2001). The Hydracarids were determined using the dichotomous keys of Cook (1974), Smith (1976), Smit (2020), Perrier (1979), and Krantz (1975; 1978). The determination of Odonata larvae is facilitated by the use of three thematic works (Gerken and Sternberg, 1999; Heidemann and Seidenbusch, 2002; Cham, 2007). For mayflies, systematic identification was carried out using the keys of Belfiore (1983), Gattolliat and Sartori, 2008 and Gattolliat et al, (2023). Generic and specific diagnoses of copepod specimens were carried out using the Perrier (1979) key. Determinations of aquatic Heteroptera were made using the work of Poisson (1957), Jansson (1969; 1986), and Dethier (1986). The Cladocerans were identified using the dichotomous key of Danielopol (1965), and Amoros (1984). Finally, the Ostracods were carried out using the key of Meisch (2000).

2.3 WATER PHYSICOCHEMICAL ANALYSES

Sampling and measurements of physicochemical parameters of water were performed seasonally out over one year in three selected sites at a depth of 30 cm. Water samples were taken in sterile plastic bottles of 1L capacity and stored at 4 °C (Aminot and Chaussepied, 1983). Physicochemical measurements: Temperature (T° C), pH, and dissolved oxygen (O₂ mg/l) were carried out in situ using a multi-parameter analyzer (WTW Multi 340 i). While nitrate: NO₃⁻, nitrite:

NO_2^- , ammonium: NH_4^+ and orthophosphate: PO_4^{3-} were determined by spectrophotometric technique according to the methods described by the American Public Health Association (APHA, 1999). The concentrations (mg/l) of elements measured were calculated based on standard curves established by Aminot and K  rouel (2004).

2.4 DIVERSITY PARAMETERS

The biodiversity of the aquatic macroinvertebrate assemblage was studied seasonally. Several parameters of biodiversity were used: The relative abundance (RA), which represents the ratio between the number of individuals and the total number of individuals (N), and the mean abundance (N_m) is the average number of individuals per species. Occurrence (Occ) was calculated for each species by the number of times the species was found / the total number of monthly surveys. Four classes of occurrence were distinguished: very accidental species (Vac) with an occurrence less than 10%, accidental species (Acc) with occurrences between [10-25%], common species (Cmn) with [25-50%] of the samples, and constant species (Cst) with an occurrence equal to or greater than 50% (Bigot and Bodot, 1973). The aquatic macroinvertebrate fauna diversity structure is assessed by calculating the Shannon-Weaver diversity index H' (Shannon and Weaver, 1949), the Simpson's diversity index (D) (Simpson, 1949), and Pielou's evenness index J' (Pielou, 1975).

2.5 STATISTICAL ANALYSIS

Results are given as means $\pm$ SD (Standard Deviation). The statistical analyses were carried out using R, version 4.0.1 (R Core Team, 2020) with $P = 0.05$ as a significance level. We applied the non-parametric Kruskal-Wallis test followed by post hoc pairwise comparisons using Dunn's test with Holm's P-value adjustment by using the 'Dunn test' package (Dinno, 2017) to compare the medians of the different variables. The results of the spatiotemporal variation of our tested biotic and abiotic variables were represented by plotting a boxplot graph using the 'ggplot2' package (Wickham, 2016). All the statistical analyses were realized at $\alpha = 0.05$ as a significance level. For the bivariate statistical approach, we calculated the Spearman (r) correlation coefficient to account for the non-normality of some variables. The correlations were calculated two by two to highlight the relationships between the physicochemical parameters of the water and the biological parameters of the seasons. The correlation matrix was plotted using the 'CooRplot' (Kassambara, 2019). In addition, the redundancy analysis (RDA) was conducted using the 'vegan' package in R to explore the relationships between the biological variables and the physicochemical parameters. The RDA was plotted using the correlative scaling method, using RDA summary outputs, and the RDA (site x season) scores. Then, the classification trees (Hierarchical clustering dendrogram) were

drawn up to better visualize the similarities between macroinvertebrate families among the four seasons and the three sites (Dagnelie, 2013).

3 RESULTS AND DISCUSSIONS

Studying the ecology of aquatic communities involves investigating the relationships between organisms and their physical, chemical, and biological environment (El Yaagoubi et al., 2023). The composition and dynamics of aquatic populations depend on the nature and importance of ecological factors such as current speed, temperature, or oxygenation of the water, as well as other physico-chemical parameters (Lévêque, 1996; Wallace and Webster, 1996). These include aquatic macroinvertebrates, which play a key role, both in terms of biodiversity and function, in the food chain, predation, and in the production and stability of ecosystems (Mahmoudi, Bendali-Saoudi, and Soltani, 2022; Gueddouche et al., 2024). In addition, very little is known about human activities in the wetlands of the Maghreb and sub-Saharan countries (Aguiza et al., 2014; Bouchaala et al., 2019). In these biomonitoring systems, the organisms most commonly used as bioindicators are benthic macroinvertebrates (Clarke et al., 2002; Sanogo et al., 2014).

3.1 TAXONOMIC COMPOSITION, RELATIVE ABUNDANCE AND OCCURRENCE

Monthly sampling of the invertebrate fauna, spread over 12 months (March 2017 to February 2018), in the three selected sites on the periphery of Lake Tonga revealed that the aquatic macroinvertebrate community was composed of 30 species among the 42,849 individuals captured. The species belonged to 20 families (Table 1), distributed respectively among the sites as follows: 15,425 individuals at the site (S₁), 10,451 at the site (S₂), and 16,973 at the site (S₃). The Cyclopidae family occupies the first position of the population identified with a maximum relative abundance (RA=65.7%). It represents 1 species, the most abundant and best represented is *Cyclops fuscus*. The Daphnidae constitute 20% of the population with two abundant species: *Daphnia magna*, *Simocephalus vetulus*, and *C. diophthalmus*. One species identified in the family Pleidae is *Plea minutissima* with 5.03%. The Corixidae have a percentage of 3.46%, among the 2 systematically identified species, *Micronecta poweri* and *Corixa punctata* have their respective relative abundances (RA= 2.7%; 0.8%). Finally, six families identified with a relative abundance and low percentage (Arrenuridae 0.06%; Aeshnidae 0.04%; Nepidae 0.03%; Belostomatidae 0.02%; Hydrachnidae 0.02%; Microveliinae 0.007%).

The identified invertebrate stand presented a total richness of 30 species and respective species richness at the sites: 30 species belonging to site 1; 29 species at site 2 and 27 species at site 3. However, it can be noted that the ranking of the species according to the index of relative

abundance, shows that the species *C. fuscus* is in the first position with 28,161 individuals (RA= 65.7% higher value). In the second position is *D. magna* with 4,478 individuals (RA= 10.45%) and in the third position *S. vetulus* with 2,411 individuals (RA= 5.63%). While the rest of the species had low relative abundances (RA < 5%). Seasonal structural analysis of the population shows a high abundance of 28,435 specimens collected and 27 species during winter. On the other hand, the lowest abundance was collected during the autumn with 221 individuals and 13 species, while spring (13,251 individuals) and summer (942 individuals) indicated intermediate abundance (Table 1).

Table 01: Systematic list, the abundance of aquatic macroinvertebrate species sampled in Tonga lake, Algeria (2017 - 2018), and frequency of occurrence notes. RA, relative abundance (%); C, class; N, the total number of individuals caught; VC, very accidental species; AC, accidental species; CM, common species; CN, constant species; Occ, frequency of occurrence.

Family [RA%]	Species	Spring				Summer				Autumn				Winter				Overall			
		N	RA	Occ	Sc	N	RA	Occ	Sc	N	RA	Occ	Sc	N	RA	Occ	Sc	N	RA	Occ	Sc
Culicidae [0.6%] Meigen 1860	<i>Culex pipiens</i> Linnaeus, 1758	23	0.05	100	CN	24	0.05	88.9	CN	—	—	—	—	19	0.04	77.8	CN	66	0.15	66.66	CN
	<i>Culex martinii</i> Medschid, 1930	27	0.06	100	CN	27	0.06	77.8	CN	—	—	—	—	1	0.002	11.1	AC	55	0.13	47.22	CM
	<i>Culex thileri</i> Theobald, 1903	—	—	—	—	5	0.01	44.4	CM	2	0.004	11.1	AC	2	0.004	11.1	AC	9	0.02	16.66	AC
	<i>Anopheles maculipennis</i> Meigen, 1818	71	0.16	100	CN	41	0.09	100	CN	—	—	—	—	5	0.01	22.2	AC	117	0.27	55.55	CN
Corixidae [3.46%] Leach 1815	<i>Corixa punctata</i> (Illger, 1807)	149	0.35	100	CN	47	0.11	100	CN	7	0.02	55.6	CN	131	0.3	100	CN	334	0.78	88.88	CN
	<i>Micronecta poweri</i> (Douglas & Scott 1869)	582	1.36	100	CN	66	0.15	100	CN	20	0.05	77.8	CN	480	1.12	100	CN	1148	2.7	94.44	CN
Microveliinae [0.007%] China & Usinger, 1949	<i>Microvelia gracillima</i> Reuter, 1883	3	0.007	22.2	AC	—	—	—	—	—	—	—	—	—	—	—	—	3	0.007	5.55	VC
Belostomatidae [0.02%] Leach, 1815	<i>Hydrocyrius colombiae</i> Spinola, 1850	—	—	—	—	7	0.02	33.3	CM	—	—	—	—	—	—	—	—	7	0.02	8.33	VC
Naucoridae [1.2%] Fallen 1814	<i>Ilyochoris cimicoides</i> (Linnaeus, 1758)	302	0.70	100	CN	62	0.14	100	CN	32	0.07	77.8	CN	110	0.26	100	CN	506	1.2	94.44	CN
Nepidae [0.03%] Linnaeus 1775	<i>Nepa cinerea</i> Linnaeus, 1758	1	0.002	11.1	AC	3	0.007	22.2	AC	4	0.009	33.3	CM	4	0.009	22.2	AC	12	0.03	22.22	AC
Notonectidae [0.6%]	<i>Anisops sardea</i>	53	0.12	88.9	CN	9	0.02	44.4	CM	—	—	—	—	9	0.02	55.6	CN	71	0.16	47.22	CM

Latreille 1802	Herrich-Schaeffer, 1849																				
	<i>Notonecta glauca</i> Linnaeus, 1758	131	0.30	100	CN	13	0.03	55.6	CN	11	0.025	66.7	CN	21	0.05	88.9	CN	176	0.41	77.77	CN
Pleidae [5.03%] Fieber 1851	<i>Plea minutissima</i> Leach, 1817	###	2.62	100	CN	106	0.25	100	CN	##	0.25	100	CN	817	1.9	100	CN	2156	5.03	100	CN
Eylaidae [0.22%] Leach 1815	<i>Eylais hamata</i> Koenike, 1897	39	0.09	100	CN	3	0.007	22.2	AC	—	—	—	—	52	0.12	77.8	CN	94	0.22	50	CM
Pionidae [0.33%] Thor1900	<i>Piona uncata</i> (Koenike, 1888)	48	0.11	100	CN	14	0.03	66.7	CN	4	0.009	22.2	AC	60	0.14	100	CN	126	0.3	72.22	CN
	<i>Forelia onondaga</i> Habeeb, 1966	4	0.009	33.3	CM	3	0.007	11.1	AC	—	—	—	—	11	0.025	44.4	CM	18	0.04	22.22	AC
Arrenuridae [0.06%] Thor 1900	<i>Arrenurus balladoniensis</i> Halik, 1940	10	0.02	77.8	CN	9	0.02	55.6	CN	—	—	—	—	8	0.019	44.4	CM	27	0.06	44.44	CM
Hydrachnidae [0.02%] O.F. Müller 1776	<i>Hydrachna cruenta</i> Müller, 1776	—	—	—	—	3	0.007	11.1	AC	—	—	—	—	6	0.01	33.3	CM	9	0.02	11.11	AC
Daphnidae [20%] Straus 1820	<i>Daphnia magna</i> Straus, 1820	###	4.55	100	CN	50	0.17	88.9	CN	20	0.05	77.8	CN	2460	5.74	100	CN	4478	10.45	91.66	CN
	<i>Simocephalus vetulus</i> (Müller, 1776)	###	2.6	100	CN	—	—	—	—	3	0.007	11.1	AC	1298	3.03	100	CN	2411	5.62	52.77	CN
	<i>Simociphalus expinosus</i> (Koch, 1841)	501	1.17	100	CN	—	—	—	—	—	—	—	—	1171	2.73	100	CN	1672	3.9	50	CM
Cyclopidae [65.7%] Dana 1846	<i>Cyclops fuscus</i> (Jurine, 1820)	###	15.8	100	CN	109	0.25	77.8	CN	—	—	—	—	21299	49.7	100	CN	28165	65.7	69.44	CN
Cyprididae [0.3%] Baird 1845	<i>Cypris bispinosa</i> Lucas, 1849	36	0.08	88.9	CN	7	0.02	44.4	CM	—	—	—	—	24	0.06	66.7	CN	67	0.15	50	CM
	<i>Eucypris virens</i> (Jurine, 1820)	34	0.08	100	CN	5	0.01	22.2	AC	—	—	—	—	19	0.04	55.6	CN	58	0.13	44.44	CM
Baetidae [1.7%] Leach 1815	<i>Baetis pavidus</i> Grandi, 1949	95	0.22	100	CN	101	0.23	100	CN	—	—	—	—	134	0.3	77.8	CN	330	0.77	69.44	CN
	<i>Cloeon dipterum</i> (Linnaeus, 1760)	85	0.2	100	CN	102	0.24	100	CN	6	0.01	33.3	CM	207	0.5	100	CN	400	0.93	83.33	CN
Leptophlebiidae [0.3%] Banks 1900	<i>Choroterpes atlas</i> Soldan & Thomas, 1983	22	0.05	77.8	CN	30	0.07	88.9	CN	—	—	—	—	80	0.19	100	CN	132	0.3	66.66	CN
Libellulidae [0.23%] Rambur 1842	<i>Crocotothemis erythrae</i> (Brullé, 1832)	61	0.14	77.8	CN	32	0.07	66.7	CN	2	0.004	11.1	AC	3	0.007	33.3	CM	98	0.23	47.22	CM

Aeshnidae [0.04%] Rambur 1842	<i>Anax imperator</i> Leach, 1815	19	0.04	66.7	CN	1	0.002	11.1	AC	—	—	—	—	—	—	—	—	20	0.04	19.44	AC
Lestidae [0.2%] Calvert 1901	<i>Lestes virens</i> (Charpentier, 1825)	14	0.03	66.7	CN	63	0.15	77.8	CN	3	0.007	33.3	CM	4	0.009	33.3	CM	84	0.19	52.77	CN

Source: Prepared by the authors (2024).

The region in which we are working has a Mediterranean climate, with mild, rainy winters and hot, dry, sunny summers from June to November. The macroinvertebrates collected belong to twenty families: Culicidae, Corexidae, Microveliinae, Belostomatidae, Naucoridae, Nepidae, Notonectidae, Pleidae, Eylaidae, Pionidae, Arrenuridae, Hydrachnidae, Daphnidae, Cyclopidae, Cyprididae, Baetidae, Leptophlebiidae, Libellulidae, Aeshnidae and Lestidae. The order Heteroptera is fairly well represented, with 7 taxa, followed by Hydracaria (6 taxa) and Odonata (3 taxa). The Heteroptera family is the most diverse (9 species). In general terms, the faunal composition of the aquatic macrofauna of Lake Tonga corresponds to that of African freshwater (Dejoux, Lauzanne, and Lévêque, 1969; Durand and Lévêque, 1981; Diomandé et al., 2009). However, the Culicidae, Cladocera, and Ephemera are fairly well represented in terms of both species and families. The relatively diverse presence of Ephemeroptera (3 taxa belonging to 2 families) reflects good water quality. These organisms are known to live in well-oxygenated environments of good quality (Enkhnasan and Boldgiv, 2020). Research carried out in India has also shown that the degradation in water quality, decrease in the dominance of Ephemeroptera, Plecoptera, and Trichoptera, and an increase in Annelids and Chrimonoids are the key factors that reflect anthropogenic stress on the Baldi River (Singh et al., 2022). According to the results obtained from all sites, site 3 is the most abundant with 16 973 individuals. The abundance at the other two sites was relatively lower, at 15,425 and 10,451 individuals respectively. This can also be explained by the development of aquatic vegetation, which provides shelter for the development of pre-imaginal stages and is a source of food for phytophagous species. The other sites are located close to urban areas and on the banks of which seasonal agriculture is practiced, with intensive pumping of water from the lake. In terms of time, the winter (wet season) is the most fruitful period, with a total of 28,435 individuals, while during the dry period (summer), the banks are very cracked. Of the 20 families of fauna identified, 18 families are common to all 3 sites. However, the Helophoridae and Microveliinae families are only found at site S₁. The Hydrachnidae family is only present at sites 1 and 2.

The composition of the Culicidae population reveals the existence of two sub-families: the *Culicinae*, represented by the genus *Culex*, which occupies first place and is represented by 3

species. Our study reveals that the genus *Culex* is dominant with two species (*Cx. pipiens*, *Cx. theileri*). However, *Cx. pipiens* is the best-represented species, which is consistent with the results of Boudemag, Bendali-Saoudi, and Soltani (2013); Trari and Dakki (2017); Benmalek et al. (2018). This species shows plasticity at the level of all the deposits surveyed in Algeria and has been encountered by several authors in the same types of deposits: fresh and pure water, and overgrown with vegetation (Brunhes et al., 1999; Amara Korba et al., 2016; Alayat et al., 2023). On the other hand, the subfamily Anophelinae, with *An. maculipennis* the most abundant and most frequently collected species. Depending on the season, spring showed significant abundance, with 121 individuals and three species. Autumn showed the lowest numbers of *An. Maculipennis*. This species has been the subject of several studies: since Guy (1959); Senevets and Andarelli (1960), and Alayat et al. (2023) who report the presence of five species of *Anopheles*. The results obtained by Boudemag et al. (2013) and Serradj, Bendali-Saoudi, and Soltani (2018) confirmed the presence of *Anopheles maculipennis sacharovi* in Collo and Lake of Birds. This complex species is of medical and veterinary interest, as it is the main vector of malaria in North Africa.

The abundance of aquatic heteropterans collected was 4,413 individuals distributed between the sites. Nine species were identified, belonging to 7 families. In a study carried out by Annani, Alfarhan, and Samraoui (2012) on Algerian wetlands in the Northeast, particularly in Constantine and Setif Provinces. They identified 35 species; six species are common with those in our study: *H. columbiae*; *N. cinerea*; *C. punctata*; *N. glauca*; *A. sardea*. Four of the species identified were cited by Serradj et al. (2018) in the Lake of Birds (north-east Algeria): *N. cinerea*; *C. punctata*; *N. glauca*; *A. sardea*. The same species were observed in Morocco by L'mohdi et al. (2008).

The hydracarian fauna was rich, with 5 species belonging to four families (Pionidae: genera *Piona* and *Forelia*, Hydrachnidae: *Hydrachna*; Eylaidae: *Eylais*; Arrenuridae: *Arrenurus*). The Pionidae family predominates at most sites. In Collo (north-east Algeria), five species have been identified. These species belong to three families: Eylaidae (*E. hamata*, *E. galeata*); Hydrachnidae (*Hydrachna murati* and *Hydrachna globosa*) and Hydryphantidae (*Eupatra rotunda*) (Boudemag, Bendali-Saoudi, and Soltani, 2018). The species *P. uncata* has the highest density, is the most widespread and has already been the subject of a morphometric study of females (Bendali-Saoudi, Gacem, and Soltani, 2013). The species *E. hamata* is the most abundant in Lake of Birds (Serradj et al., 2018) and has also been found in Lac Tonga (Bendali-Saoudi, Gacem, and Soltani, 2014). Finally, the family Hydrachnidae with the genus *Hydrachna* has also been reported from the same site (Bendali-Saoudi et al., 2014; Khedimallah and Tadjine, 2016).

For the Ephemeroptera group, 862 individuals were collected, divided into 2 families (Baetidae, Leptophlebiidae) and 3 species. The Baetidae family is the most diverse and is

represented by 2 species: *B. pavidus* and *C. dipterum*. A Mediterranean species *C. dipterum*, frequents the main hydrographic networks of the Atlas Gattolliat et al. (2023). In Tunisia, it is the most frequent and abundant species (Boumaiza and Thomas, 1995). In the southern Ivoir Coast, the Baetidae family has also been reported by Yapo, Atsé, and Kouassi (2015) in fishponds. In addition, the Family Leptophlebiidae, represented by a single species: *C. atlas* is endemic to the Maghreb. In a study conducted by Yalles-Satha et al. (2020), on the 17 sites distributed along an altitudinal gradient in two permanent Wadis in northern Algeria, they identified eight species belonging to six families; three species are common with that of our study. In Morocco, the three identified species were also cited by Abessolo, Khebiza, and Messouli (2020) in the Ourika watershed (High Atlas), and by El Alami, El Yaagoubi et al. (2022) in various Moroccan areas.

Odonate larvae, which have an aquatic lifestyle, are a prime group for the assessment, monitoring, and management of wetlands according to Mairif, Bendifallah, and Doumandji (2023). The results obtained allowed us to report the presence of three species (202 individuals) belonging to 3 families (Libellulidae, Aeshnidae, and Lestidae). In Numidia, where the El-Kala region is located and which has a rich palette of wetlands exclusive to the Mediterranean basin, is home to exceptional odonatological biodiversity (Samraoui and Belair, 1998): a total of 45 species of odonata ($\sim \frac{3}{4}$ of Algerian species) have been recorded (Samraoui and Corbet, 2000). Two identified species *C. erythrae* and *A. imperator* were cited by El Bouhissi, Chaib, Houhamdi, and Khelifa (2022) in Sidi Bel-Abbes (Northwest Algeria). In particular, the species *A. imperator* and *L. virens* are also reported in the Akfadou massif forest in Algeria by Chelli and Moulai (2019).

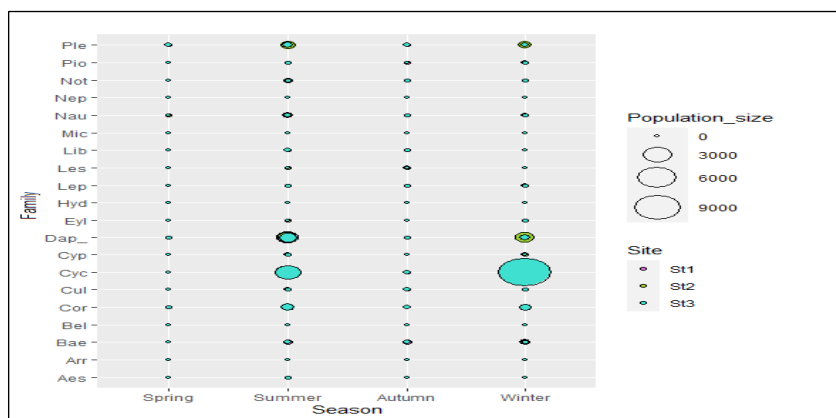
The macrobenthic crustacean fauna is represented by 6 species, belonging to three families: Daphnidae with three species; Cyclopidae with one species; Cyprididae with two species. Four of the species identified were cited by Houmani et al. (2023) at Lake Echatt (north-east Algeria): *D. magna*; *S. vetulus*; *S. expinosus*; *C. fuscus*. In Tunisia Marrone et al. (2020) reported *E. virens* was the most frequently encountered ostracod. The results of Menail, Scharf, and Amarouayache (2023) study shows that in the Algerian Sahara and High Plains: 24 ostracod species were identified; belonging to four families (Darwinulidae, Ilyocyprididae, Cyprididae and, Cytherideidae); the Cyprididae family is common with that of our study. The work carried out by Belala, Hadjab et al. (2020) found 22 taxa (families) and 4,064 individuals, and is dominated by Insects (51%) and crustaceans (28%); this is consistent with our survey.

3.2 SEASONAL OCCURRENCE AND PHENOLOGY OF SPECIES

The analysis of the occurrence and spatial phenology of 30 species inventoried in Tonga Lake indicates the following characteristics according to (Figure 2). Two species were classified as

accidental: *Microvelia gracillima* and *Hydrocyrius colombiae*, Acc (FO < 10%). Five species ranked rare: *Nepa cinerea* and *Culex thieleri*; *Forelia onondaga*; *Hydrachna cruenta*; *Anax imperator*, Rar (25% > FO ≥ 10%), eight species were ranked casual: *Culex martinii*, *Anisops sardea* and *Eylais hamata*; *Arrenurus balladoniensis*; *Simociphalus expinosus* and *Cypris bispinosa*; *Eucypris virens*; *Crocothemis erythrae*, Occ (50% > FO ≥ 25%). Eight species were ranked common: *Culex pipiens*; *Anopheles maculipennis*; *Piona uncata*; *Lestes virens*; *Simocephalus vetulus*, *Cyclops fuscus*; *Baetis pavidus*, and *Choroterpes atlas* Cmn (75% > FO ≥ 50%). The following seven species are ranked as frequent: *Corixa punctata*; *Micronecta poweri*; *Ilyochoris cimicoides*; *Notonecta glauca*; *Plea minutissima* *Daphnia magna*; and *Cloeon dipterum*, Frq (FO ≥ 75). The presence of each family varies from season to season at the study sites. In summer, when the maximum number of families was reported (19 families), nine families were constant (Cyclopidae, Corixidae, Naucoridae, Notonectidae, Pleidae, Daphnidae, Baetidae, Leptophlebiidae, and Lestidae), six were common (Culicidae, Eylaidae, Arrenuridae, Pionidae, Cyprididae, and Libellulidae), three were accidental (Nepidae in the four seasons; Hydrachnidae in summer and winter, Aeshnidae in spring). Two families were classified as very accidental: Microveliinae during spring as well as Belostomatidae in the summer (Table 1, Figure 2).

Figure. 2: Phenology of abundances of aquatic macroinvertebrate families sampled in the Tonga (March 2017 – February 2018).



Source: Prepared by the authors (2024).

3.3 SEASONAL AND SPATIAL PATTERNS OF DIVERSITY INDICES

The results of the diversity and homogeneity indices are shown in Table 2. Indeed, the maximum richness was observed in summer (S = 19 families) with an abundance of 942 individuals, while the other indices present maximum values in spring (H' = 1.439, D = 0.656 (Simpson's Index), J' = 0.227) with an abundance of 13,251 individuals. Regarding the specific spatial richness, the highest values of the ecological indices are recorded in site 1 (S = 20 species; H' = 1.284 and D =

0.554, N = 15,425 individuals), except of the abundance, whose maximum value is recorded at site 3 (N = 16 973 individuals, J' = 0.098).

The kinetic abundance of the different families and the distribution of species change according to the season and the site. Their abundance varies according to environmental factors and is linked to the different annual ecological conditions and climatic conditions: it decreases in autumn, with winter and spring representing maximum abundance. The El Kala region has proved to be a veritable "pocket" of relictual species (Samraoui et al., 2021).

Table 02: Spatio-temporal evolution of biodiversity indices of macroinvertebrate families sampled in the Tonga lake (March 2017 - February 2018). S: species richness, H': Shannon Weaver index, D: Simpson's diversity index, SRI: Simpson's reciprocal index, J': Pielou's evenness index, N: the total number of individuals).

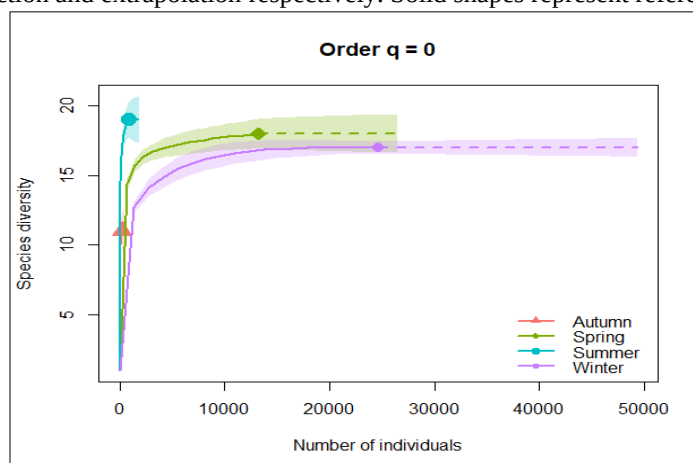
Factor	S	N/S	H'	D	SRI	SRI/S	J'	N
Spring	18	736.16	1.439	0.656	1.523	0.084	0.227	13251
Summer	19	49.6	2.404	0.886	1.128	0.059	0.301	942
Autumn	11	20	1.659	0.714	1.399	0.127	0.298	221
Winter	17	1672.64	0.692	0.280	3.559	0.209	0.099	28435
St_1	20	771.25	1.284	0.554	1.803	0.090	0.185	15 425
St_2	19	550	1.354	0.609	1.642	0.086	0.207	10 451
St_3	18	942.94	0.707	0.283	3.529	0.196	0.098	16 973

Source: Prepared by the authors (2024).

3.4 INTERPOLATION OF MACROINVERTEBRATE SPECIES RICHNESS

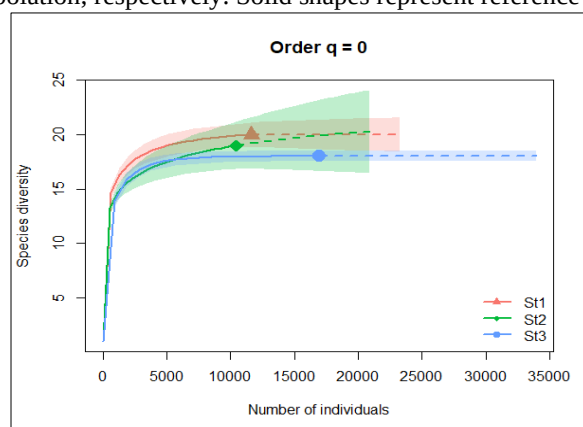
Seasonal analysis of richness curves as a function of the number of individuals observed (Figure 3). The diversity is fairly close between the four seasons. In particular, summer, spring, and winter show comparable diversity; they have populations of the macroinvertebrate significantly more diverse than the autumn season. Having recorded a specific richness comparable to the other season, autumn presents a lower diversity; it provides a lower number of species and individuals. With regard to the spatial study concerned, richness curves according to the number of individuals observed (Figure 4) indicate that site 3 presents the highest value, followed by site 1, then site 2. The hierarchical clustering analysis of macroinvertebrate families based on Euclidean peer groups (UPGMA) revealed three discriminating groups: Group 1 included the autumn (dry season), Group 2 included the spring season at all three sites, in addition to autumn (site 1) as an exception because it is close to the center of the lake filled with water during the dry season, giving it the characteristic of a microclimate that resembles the spring season, and Group 3 included the remaining seasons (winter and summer) at all sites (Figure 5).

Figure. 3: Individual seasonal rarefaction and extrapolation (R/E) curves with 95% confidence intervals (shaded areas) for the macroinvertebrate community as a function of diversity order $q = 0$ (species richness, Chao1), solid and dashed segments represent rarefaction and extrapolation respectively. Solid shapes represent reference samples.



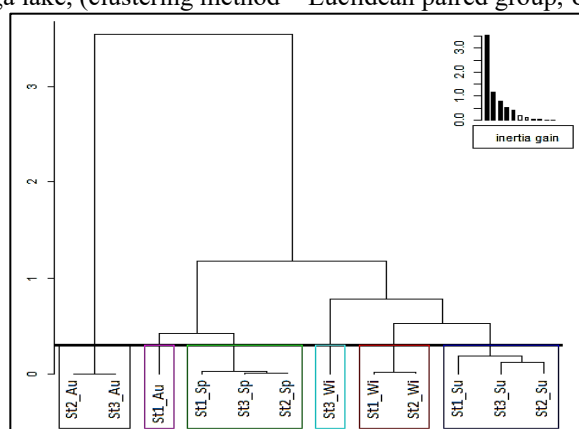
Source: Prepared by the authors (2024).

Figure. 4: Individual spatial rarefaction and extrapolation (R/E) curves with 95% confidence intervals (shaded areas) for the macroinvertebrate community as a function of diversity order $q = 0$ (species richness). Solid and dashed segments represent rarefaction and extrapolation, respectively. Solid shapes represent reference samples.



Source: Prepared by the authors (2024).

Figure 5. Hierarchical clustering dendrogram illustrating abundance-based similarity of macroinvertebrate families among the four seasons in Tonga lake, (clustering method = Euclidean paired group, UPGMA).



Source: Prepared by the authors (2024).

3.5 SPATIOTEMPORAL VARIATIONS AND EFFECTS OF ENVIRONMENTAL FACTORS

Following normality tests, the difference between the means of inter-site and inter-season comparisons of physicochemical and macroinvertebrate families (07 variables) were verified by a non-parametric test (Kruskal Wallis test). Thus, the spatio-temporal trends of almost all the water parameters measured revealed non-significant variations. However, one exception was a variation between seasons, with a single significant difference in température ($P < 0.05$) (Table 3).

Table 03: Inter-site and inter-season comparison of physico-chemical parameters (7 variables).

Variables	Factor			
	Sites		Season	
	P value	Observation	P value	Observation
Temperature (T°C)	0.810	Ns	0.035	*
pH	0.645	Ns	0.069	Ns
Dissolved oxygen O ₂	0.316	Ns	0.647	Ns
Nitrate (NO ₃ ⁻)	0.841	Ns	0.097	Ns
Nitrite (NO ₂ ⁻)	0.897	Ns	0.397	Ns
Ammonium (NH ₄ ⁺)	0.704	Ns	0.110	Ns
Orthophosphate (PO ₄ ³⁻)	0.815	Ns	0.060	Ns

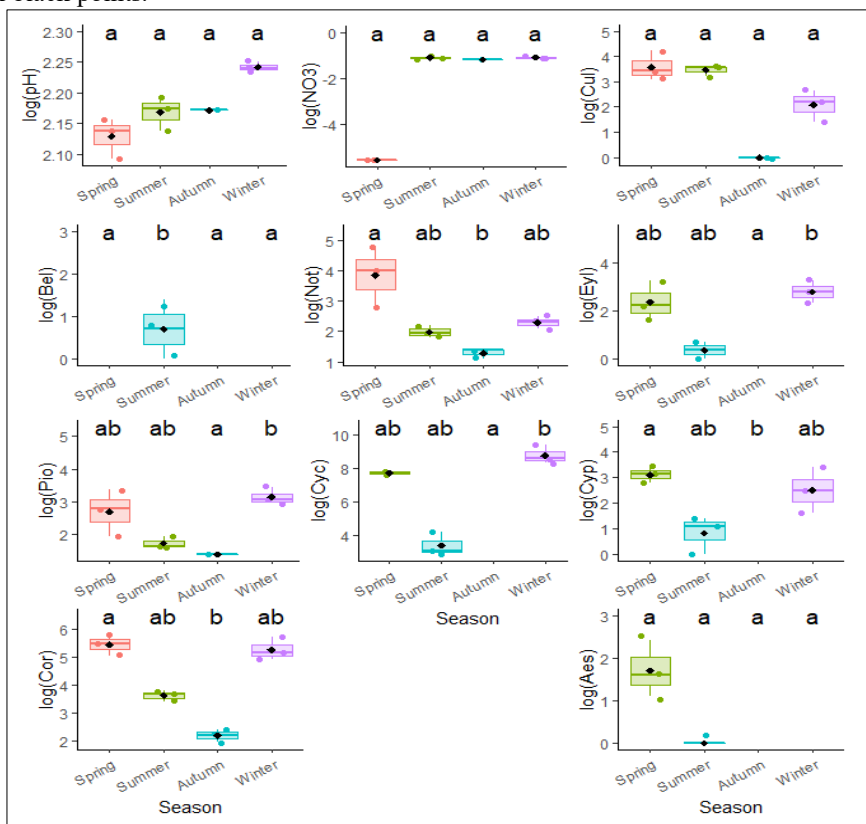
* ($P \leq 0.05$), Ns: not significant.

Source: Prepared by the authors (2024).

3.6 SEASONAL VARIATIONS IN PHYSICOCHEMICAL AND BIOLOGICAL PARAMETERS

The comparison between seasons by using the nonparametric Kruskal Wallis test revealed significant differences ($P < 0.05$) for the following variables: pH, NO₃⁻, Culicidae, Belostomatidae, Notonectidae, Eylaidae, Pionidae, Cyclopidae, and Cyprididae. In addition, the post hoc pairwise comparisons of Dunn's tests are presented by boxplots (Figure 6). The pairwise comparison with the Tukey test indicates that: Boxplots expressing seasonal variations of abiotic and biotic parameters in Tonga Lake show clear homogeneity (non-significant differences). Four homogeneous groups are reported for pH, NO₃⁻, Culicidae, and Aeshnidae, so there is no significant difference between the four seasons. The boxplot analysis of the variations in seasons of the six families (Notonectidae, Eylaidae, Pionidae, Cyclopidae, Cyprididae, Corexidae) shows the existence of a clear heterogeneity, indicating a very highly significant difference ($P = 0.000$). On the other hand, there is one heterogeneous group only in the summer season for the family of Belostomatidae; the difference is significant (Figure 6).

Figure. 6: Boxplots displaying seasonal variations of physicochemical and biological parameters in Tonga Lake. pH: Hydrogen potential; NO_3^- : Nitrate; Culicidae: Cul; Belostomatidae: Bel; Notonectidae: Not; Eylaidae: Eyl; Pionidae: Pio; Cyclopidae: Cyc; Cyprididae: Cyp; Corexidae: Cor; Aeshnidae: Aes. Different letters indicate a significant difference among levels of variables ($p < 0.05$), and the same letters are significantly non different. Central box limits show the interquartile range (IQR) with the first (lower box bound) and third (upper box bound) quartiles. Outliers are indicated by small black points.

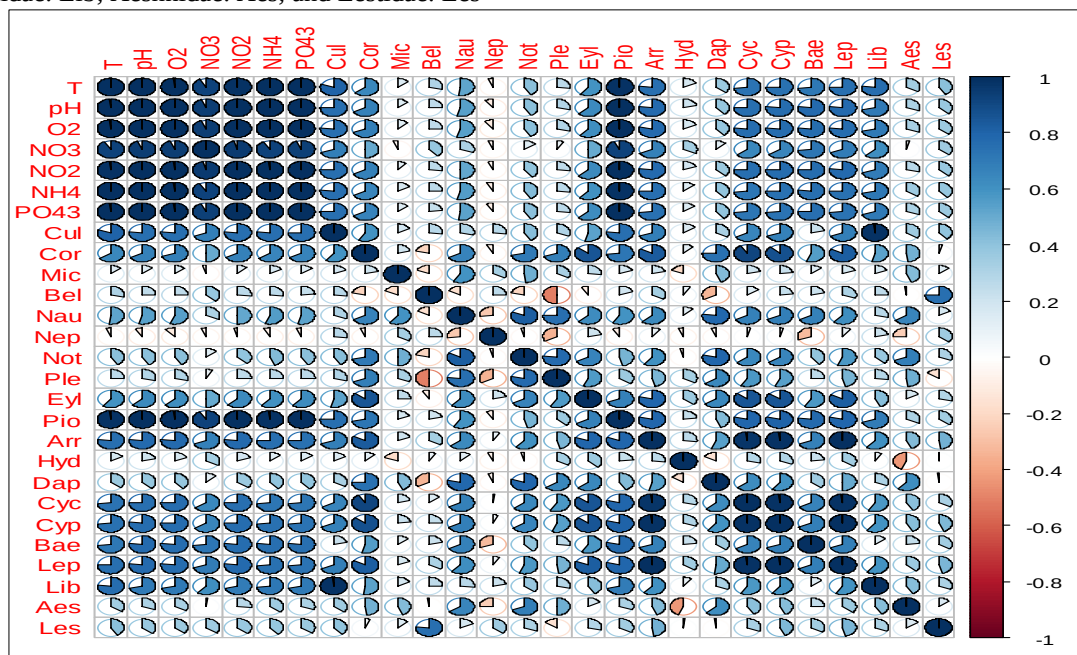


Source: Prepared by the authors (2024).

3.7 CORRELATION ANALYSIS BETWEEN ENVIRONMENTAL PARAMETERS

Out of 378 correlation tests between physicochemical parameters of water and macroinvertebrate families, 359 were statistically significant ($P < 0.05$). Most of these significant correlations were positive and were established between T°C , pH, O_2 , NO_3^- , Nitrate, NO_2^- , Nitrite, NH_4^+ , PO_4^{3-} , Culicidae, Eylaidae, Pionidae, Arrenuridae, Cyclopidae, Leptophlebiidae, Libellulidae, Lestidae, and other parameters. Correlations indicated that Belostomatidae was negatively related to Naucoridae, Notonectidae, Pleidae, Eylaidae, and Daphnidae. Besides, Nepidae was negatively correlated with Notonectidae, Pleidae, Pionidae, Hydrachnidae, and Daphnidae (Figure 7).

Figure. 7: Corrplot of the Spearman correlation calculated between all the physico-chemical parameters analysed and the species inventoried. T: Temperature; pH: Hydrogen potential; O₂: Dissolved oxygen, NO₃⁻: Nitrate; NO₂⁻: Nitrite; NH₄⁺: Ammonium; PO₄³⁻: Orthophosphate; Culicidae: Cul; Corexidae: Cor; Microveliinae: Mic; Belostomatidae: Bel; Naucoridae: Nau; Nepidae: Nep; Notonectidae: Not; Pleidae: Ple; Eylaidae: Eyl; Pionidae: Pio; Arrenuridae: Arr; Hydrachnidae: Hyd; Daphnidae: Dap; Cyclopidae: Cyc; Cyprididae: Cyp; Baetidae: Bae; Leptophlebiidae: Lep; Libellulidae: Lib; Aeshnidae: Aes; and Lestidae: Les

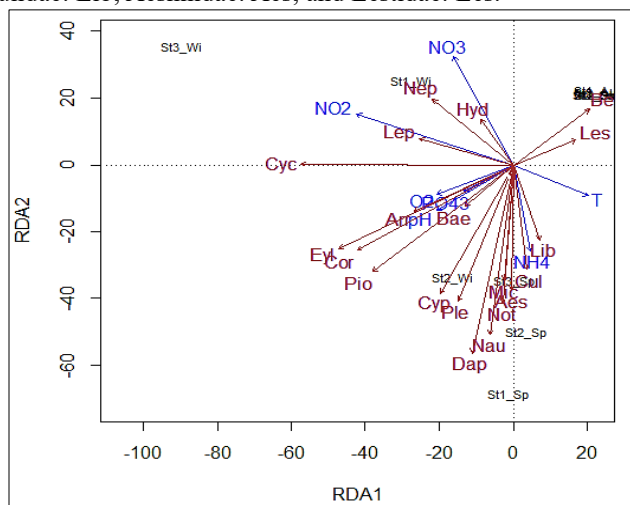


Source: Prepared by the authors (2024).

3.8 FRESHWATER MACROINVERTEBRATE FAMILIES' RELATIONSHIPS

Redundancy analysis (RDA) (Figure 8) explores the relationships between the diversity of aquatic macroinvertebrate families and the physicochemical variables in water. According to the first axis, all water parameters were negatively correlated with NH₄⁺ and temperature, which were positioned on the negative side of the axis with most families (Hydrachnidae, Nepidae, Leptophlebiidae, Cyclopidae, Baetidae, Arrenuridae; Eylaidae, Corexidae, Pionidae, Cyprididae, Pleidae, Daphnidae, Naucoridae, Notonectidae, Aeshnidae, and Microveliinae). The families Lib and Cul are placed on the positive side of the same axis, so it is positively correlated with NH₄⁺ and temperature. According to the second axis, Libellulidae, and Culicidae are positively correlated with NH₄⁺, but negatively correlated with NO₂⁻ and NO₃⁻. However, Hydrachnidae, Nepidae, Leptophlebiidae, and Cyclopidae are positively correlated with water parameters: NO₂⁻ and NO₃⁻. Commonly Baetidae, Arrenuridae, Eylaidae, Corexidae, and Pionidae were positively associated with pH, O₂, and PO₄³⁻. Then, six families were negatively correlated with all water parameters: Belostomatidae, Lestidae, Cyprididae, Pleidae, Naucoridae, and Daphnidae.

Figure. 8: Redundancy analysis diagram displaying relationships between water physicochemical parameters and macroinvertebrate families in Tonga Lake. T: Temperature; pH: Hydrogen potential; O₂: Dissolved oxygen, NO₃⁻: Nitrate; NO₂⁻: Nitrite; NH₄⁺: Ammonium; PO₄³⁻: Orthophosphate; Culicidae: Cul; Corexidae: Cor; Microveliinae: Mic; Belostomatidae: Bel; Naucoridae: Nau; Nepidae: Nep; Notonectidae: Not; Pleidae: Ple; Eylaidae: Eyl; Pionidae: Pio; Arrenuridae: Arr; Hydrachnidae: Hyd; Daphnidae: Dap; Cyclopidae: Cyc; Cyprididae: Cyp; Baetidae: Bae; Leptophlebiidae: Lep; Libellulidae: Lib; Aeshnidae: Aes; and Lestidae: Les.



Source: Prepared by the authors (2024).

The physicochemical parameters of water can play an important role both in the biology of a species and in the structure and dynamics of the whole biocenosis (Alayat et al., 2023; Houmani et al., 2023). The physicochemical parameters of the water were: temperature, pH, dissolved oxygen, nitrite, nitrate, ammonium, and orthophosphate. Physicochemical analysis of water at the studied sites revealed that the average abundance was lower during the autumn and winter seasons. The families identified showed no significant variation between the three sampling sites. This homogeneity in the physicochemical composition of the water is probably the result of the nature of the water source, which is almost entirely based on precipitation according to Saoudi et al. (2018). In the same context, Baaloudj et al. (2020) reported that the Seybouse River is host to a highly diverse fauna with a wide range of vital needs, which has adapted to variations in environmental factors. According to Loucif et al. (2020), Lake Tonga is currently in a state of eutrophication and further severe degradation of the ecosystem may occur if appropriate management measures are not taken in the short term. The spatiotemporal trends of physicochemical parameters measured in the water of Lake Tonga revealed the non-significant effects of both "Site" and "Season". However, an exception was reported for the significant effect of temperature ($P < 0.05$). El Kala Park is a region with a strong agricultural vocation using chemical fertilizers and pesticides and extensive livestock farming causing the deterioration of the plant cover and also, soil contamination due to various animal discharges (Saadali et al., 2015). Spearman correlation showed that the most of correlations were positive and were established between T°C, pH, O₂, NO₃⁻, Nitrate, NO₂⁻, Nitrite, NH₄⁺, PO₄³⁻, Culicidae, Eylaidae, Pionidae, Arrenuridae, Cyclopidae, Leptophlebiidae, Libellulidae, and

Lestidae. Negative correlations were found between two families: Belostomatidae and Nepidae. Our study also points out that the lake studied is taxonomically rich with 20 families, and 30 species. Therefore, the conservation and protection of Tonga Lake are necessary (Mahmoudi et al., 2023).

4 CONCLUSION

Freshwater macroinvertebrates have been used for a long time to assess the health of water systems. However, the key factors explaining aquatic organism diversity in North African lentic ecosystems are still not addressed. This research work contributes to the realization and update of an inventory of aquatic macroinvertebrates of Lake Tonga in the El Kala National Park (Northeast Algeria). Thirty species belonging to twenty families were identified during one year. In addition, a physicochemical analysis of the water at the sampling sites was performed. For this purpose, we carried out seasonal sampling. The ecological indices showed a variability of the spatial distribution of the families according to the parameters of the water. Thus, in this study, to determine the abiotic factors influencing this distribution, the majority of families are negatively correlated with NH_4^+ and temperature. The fauna of macro-benthic invertebrates is poorly known in Algeria, as well as throughout the world. A study on biodiversity and species identification would be essential for a better understanding of this sensitive biological material, threatened by tropical action and climate change. Biological and eco-toxicological awareness must be raised by various researchers, in order to preserve protected wetlands and conserve this priceless natural wealth, so we're sounding the alarm to protect this world-renowned and classified wetland.

ACKNOWLEDGEMENTS

We would like to thank Pr Morad Bensouilah and his team (Marine and Coastal Ecobiology Laboratory (EMMAL), Badji Mokhtar University, Annaba, Algeria) for their contribution to the physicochemical analysis of the water. Special thanks to Pr. Francesco Porcelli (Laboratory DiSSiPA, Aldo Moro University of Bari, Italy) for his help in identifying some specimens. We are deeply grateful to Dr. Karima Heramza (Laboratory of Biodiversity and Pollution of Ecosystems, Chadli Bendjedid University, El Tarf, Algeria) for the statistical analysis of the data. This work was supported by the National Fund for Scientific Research of Algeria to laboratory of Applied Animal Biology (to Pr N. Soltani) and by the Ministry of Higher Education and Scientific Research of Algeria (PRFU project n° D01N01UN230120190005 to Pr F. Bendali-Saoudi).

CONFLICT OF INTEREST STATEMENT

We declare that we have no conflict of interest.

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